

Influence of the morphology of ZnO nanomaterials on photooxidation of polypropylene/ZnO composites

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ABSTRACT

In this study, the effect of zinc oxide (ZnO) morphology on photooxidation of polypropylene (PP)/ZnO nanocomposite films was analyzed by characterizations of their structure, morphology and properties. PP/ZnO nanocomposite films containing 0.25, 0.5, 0.75 and 1.0 weight percentage (wt%) of quasi-spherical (0D-ZnO) and rods (1D-ZnO) ZnO particles were prepared by melt processing followed by compression molding and they were exposed under xenon-arc light irradiation for different durations. Photooxidation process of these nanocomposites was monitored by the presence of oxidative products as well as the weight loss observed using infrared spectroscopy (FTIR) and gravimetric analysis respectively. Surface morphology of PP/ZnO nanocomposite films was analyzed by scanning electron microscopy (SEM). Molecular weight of the polymer was used to observe the extent of cleavage promoted by ZnO morphology in the polymer chains. The morphology of ZnO particles showed an important influence on photooxidation of PP/ZnO nanocomposites. A catalytic effect was observed in the photooxidation of PP nanocomposites with 0D-ZnO by an increase in the carbonyl index, weight loss and erosion of the film surface compared to the pristine PP. The 1D-ZnO nanostructures had a UV-screening effect inhibiting the photooxidation of PP and consequently the rupture of polymer chains during longer periods of time with fewer imperfections on the surface. The results obtained show the effects of the semiconductor nanomaterial concentration and morphology on photo-oxidation of polymer-semiconductor nanocomposites, in this case ZnO particles on PP/ZnO nanocomposites.

1. Introduction

Polypropylene (PP) is currently one of the most used thermoplastic polymers with a wide range of applications in packaging, automotive, electro-domestic and other industries [1]. In recent years, polypropylene nanocomposites based on 0D, 1D and 2D nanostructures have become a new way to enhance their mechanical, thermal, electrical, optical and barrier properties at low loading levels [2]. For a better understanding on thermal stability and degradation mechanisms for PP nanocomposites several studies were carried out, mainly using organoclays [3,4], TiO₂, ZnO and other nanoparticles as nanofiller [2,5,6]. The polymer nanocomposites with TiO₂ and ZnO nanoparticles have shown features dependent of their structure and morphology. TiO₂ nanoparticles with a higher content of anatase crystalline phase

promoted photodegradation of polymers such as ethylene-vinyl acetate copolymer (EVA) [7], PP [8,9], polyethylene (PE) [10], and polyvinyl chloride (PVC) [11]. ZnO nanoparticles were also used as UV screeners by incorporating them to polymer, which absorbed UV radiation and decreased the polymer photodegradation by addition of 0D-ZnO nanoparticles with surface modified organosilanes [12]. Good dispersion of ZnO is a critical parameter to obtain an efficient UV-screen effect in PP/ZnO soluble starch nanocomposites [13] or at least with good UV absorption characteristics in PP reinforced with ZnO nanorods [14]. The photocatalytic activity of ZnO towards polymer photooxidation was studied for low density polyethylene (LDPE) [15], PVC [16] and also in some biopolymers [17]. The photooxidation mechanism of LDPE nanocomposites films was explored using ZnO and TiO₂ nanoparticles resulting with a greater generation of carbonyl groups with the

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incorporation of TiO₂, while ZnO increased the formation of CO₂. Furthermore hybrid ZnO materials using matrix polymers such as PP [18], polyaniline (PANI) [19], nanocellulose [20] can be used for degradation of organic colorants with antibacterial activity. The crystalline structure, shape and size of ZnO nanoparticles were correlated with its photocatalytic activity in aqueous systems [19,21–24]. In spite of all these studies, there is still a debate on the role of ZnO during the photooxidation process in polyolefins, such as PP.

In the present work, PP/ZnO nanocomposites based on 0D and 1D ZnO particles were prepared by melt process and their nanocomposite films were obtained by compression molding. These films were exposed under xenon arc lamps during 266 h and the photooxidation of PP/ZnO films, as well as changes on the morphology were evaluated.

2. Experimental

2.1. Materials

Polypropylene homopolymer (PP) (Valtec HS013, melt flow index 11g / 10 min) was supplied by Indelpro (Altamira, Mexico). Zinc oxide (ZnO) nanoparticles (0D-ZnO) and rods (1D-ZnO) were synthesized by methods reported elsewhere. A typical process for synthesizing 0D-ZnO nanoparticles [25] is a 25 mL of 0.02 M Zn(Ac) ethanol solution was added dropwise to 25 mL of 0.04 M sodium hydroxide (NaOH) solution, over a period 25 min at 70 °C. After the complete addition of all precursors, the mixture was continuously stirred and heated for 2 h. The resulting white powder was separated by centrifugation and washed with ethanol. 1D-ZnO were also prepared following the procedure reported in literature [26]. 104 mL of a 4 M NaOH solution were added to 208 mL of deionized water under magnetic stirring, and then 104 mL of a 0.2 M Zn(NO₃)₂·6H₂O solution were added. The solution was stirred for 10 min, and then transferred to a 600 mL autoclave. The autoclave was sealed and kept at 150 °C and a pressure of 21 psi during 6 h. After complete reaction, the resulting white powder was separated from the solution by vacuum filtration and washed with deionized water and ethanol. All materials were used without further treatments.

2.2. Preparation of PP/ZnO nanocomposites

ZnO nanoparticles and PP pellets were melt-mixed in a Rondol twin screw extruder (L/D = 25:1, D = 10 mm) at 75 rpm. The temperature profile was 185 °C, 185 °C and 195 °C for each section of extruder (three sections) and 185 °C for die. The ZnO particle content in nanocomposites was 0.25, 0.50, 0.75 and 1.0 wt% respect to weight polymer (20g). A sample of unfilled PP was also processed at same conditions for comparison with PP/ZnO nanocomposite. Films of PP/ZnO nanocomposites were prepared by compression molding in a Carver press held at 3.6 MPa and 190 °C during 3 min. Films thickness was around 0.15 mm. PP/ZnO nanocomposites containing 1D-ZnO particles are referred as PP/1DZnO and PP with 0D-ZnO nanoparticles as PP/0DZnO (see Table 1).

Table 1
Samples of PP/ZnO prepared as film.

Chemical composition	Sample identification
pristine Polypropylene	PP
PP + 0.25% 0D-ZnO	PP/ 0DZnO-0.25
PP + 0.50% 0D-ZnO	PP/ 0DZnO-0.5
PP + 0.75% 0D-ZnO	PP/ 0DZnO-0.75
PP + 1.00% 0D-ZnO	PP/ 0DZnO-1.0
PP + 0.25% 1D-ZnO	PP/ 1DZnO-0.25
PP + 0.50% 1D-ZnO	PP/ 1DZnO-0.5
PP + 0.75% 1D-ZnO	PP/ 1DZnO-0.75
PP + 1.00% 1D-ZnO	PP/ 1DZnO-1.0

2.3. Photooxidation of PP/ZnO films

The films were placed into a Q-Lab xenon test chamber (Q-SUN Xe-3-HS) using three xenon arc lamps with a filter of 340 nm. The light intensity was 0.35 W/m². The film surface was continuously subjected to xenon arc radiation during 24 h up to a total exposure time of 266 h (total energy = 716.9 kJ/m²). During irradiation treatment, the temperature in black panel and air chamber was controlled at 45 °C and 40 °C, respectively. The relative humidity was controlled at 50%. The size of a film sample was about 8 cm² and 0.15 mm of thickness.

2.4. Characterization

Structural analyses of ZnO particles were performed using a X-ray diffractometer (Bruker Advanced X-Ray Solutions D8) using the K α line of Cu (1.5406 Å) in the 2 θ range of 10–70° with a step size of 0.02°. Diffuse reflectance spectra (DRS) were acquired using a UV-Vis Spectrophotometer (Lambda 35 Perkin Elmer) and used for the energy band gap determination of ZnO particles. The DRS data were transformed through a Kubelka–Munk function relating the reflectance of the sample (R_{∞}) to sample concentration [27,28]: $F(R_{\infty}) = (1 - R_{\infty})^2 / 2R_{\infty}$. Morphology of the ZnO particles was analyzed by transmission electron microscopy (TEM-FEI Titan G2 80–300) and scanning electron microscopy (SEM-FEI Nova NanoSEM 200) operated at an accelerating voltage of 5 kV. The influence of ZnO nanoparticle in photooxidation of PP film was evaluated by FTIR spectroscopy (Nicolet 6700 spectrometer) with a 4 cm⁻¹ resolution and 32 scans. Carbonyl index was calculated using the ratio of intensities between the carbonyl peak (C=O) at 1710 cm⁻¹ and a reference peak (C-H) at 2720 cm⁻¹ [12]. The control of the weight of each film was carried out during its degradation process using an analytical balance. The change in weight was determined by: change in weight = ($m_i - m_f$) / m_i where m_i is the initial weight film before irradiation and m_f is the final weight after irradiation (approximately each 24 h of exposition). The Raman spectra (Thermo Scientific DXR Raman microscope) of PP/ZnO films before xenon-arc irradiation were acquired using a 532 nm laser excitation. X-ray photoelectron spectroscopy (XPS) analysis (Thermo Scientific K-Alpha XPS) was evaluated using an instrument with monochromatized Al K α radiation ($h\nu = 1486.68$ eV). All the spectra reported were recorded with reference to C 1s peak (284.6 eV). Nonlinear fit, using Gaussian model, was performed and the full-width at half-maximum (FWHM) was maintained constant. Molecular weight of PP films was obtained in an instrument of size exclusion chromatography (Waters 150 C SEC), equipped with a differential refractometer detector and three Ultra Styragel columns having 106, 104 Å and linear mixture nominal porosity. The permeation solvent was 1,2,4-trichlorobenzene at 145 °C with an 1 mL/min of flow rate. Samples were stabilized with 2 wt% antioxidant and heated at 135 °C for 2 h to ensure complete dissolution.

3. Results and discussion

3.1. ZnO nanostructures characterization

Fig. 1a shows the SEM images for 0D-ZnO nanoparticles forming agglomerates. The nanoparticles are observed by TEM (inset of Fig. 1a) with particle size of ~ 10 nm. Fig. 1b shows well-faceted 1D-ZnO particles with prismatic or pyramidal terminations. Average particle size of the 1D-ZnO was estimated to be 11 μ m in length and 1.1 μ m in diameter, with a resulting aspect ratio (l/d) of 10. The sharpness factor ($S_f = h/b$) was determined by relating the height of the pyramidal termination to the superior flat surface of the pyramid [26], resulting in a value of 18. Also, the superior flat surface of the pyramid was of 70 nm, which agrees with the sharp and elongated terminations of the ZnO rods [26].

All ZnO samples exhibited the wurtzite crystalline phase

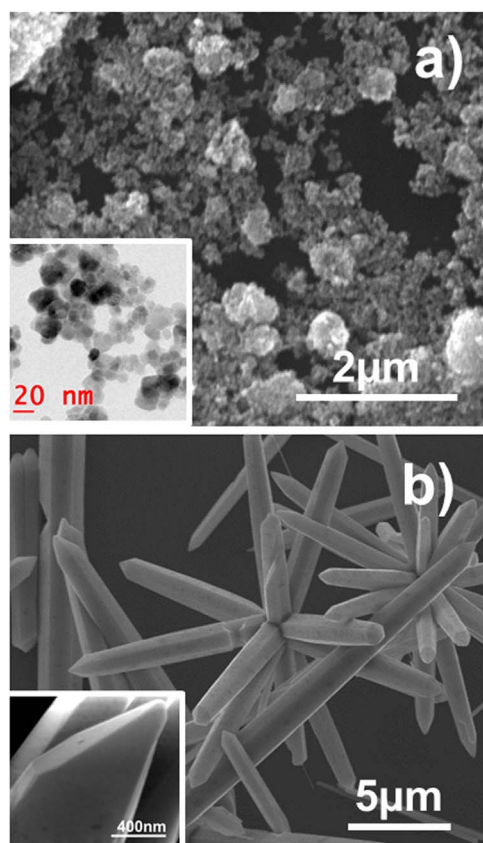


Fig. 1. SEM image of a) OD-ZnO (inset: Bright field TEM image) and b) 1D-ZnO particles.

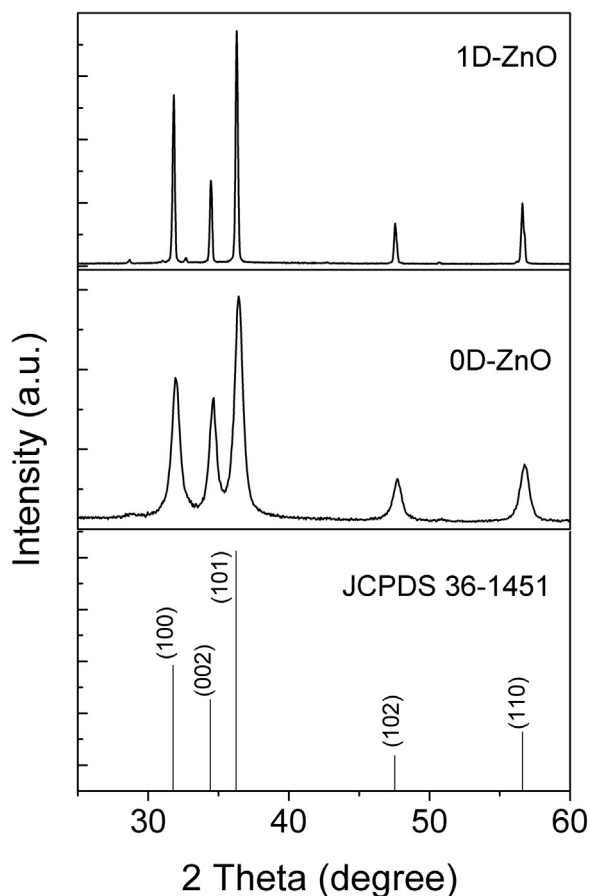


Fig. 2. XRD patterns of ZnO particles.

(hexagonal, space group $P6_3mc$). The X-ray diffraction patterns for the samples are presented in Fig. 2. All diffraction peaks were indexed using JCPDS card 36-1451 [29]. The lattice parameters were calculated from these XRD patterns resulted $a = 3.23 \text{ \AA}$ and $c = 5.17 \text{ \AA}$ for the OD-ZnO, and $a = 3.24 \text{ \AA}$ and $c = 5.20 \text{ \AA}$ for the 1D-ZnO. Such values are in agreement with the standard parameter for the wurzite crystalline structure $a = 3.24 \text{ \AA}$ and $c = 5.20 \text{ \AA}$. For 1D-ZnO, two small diffraction peaks at 2θ values of 28° and 32° are attributed to the presence of $\text{Zn}(\text{OH})_2$ (JCPDS card 20-1436) [30] as a result of incomplete transformation to ZnO [31]. No additional peaks related to impurities are detected for OD-ZnO by XRD technique. Even though both particles exhibited the same crystalline structure, the diffraction peaks of OD-ZnO are broader than those of the 1D-ZnO, indicating the nanoscale crystalline size of OD-ZnO particles (12.4 nm calculated by Scherrer's equation).

In order to determine the optical energy band gap of the ZnO particles, their diffuse reflectance spectra were obtained (Fig. 3a) using the Kubelka-Munk treatment to obtain the energy band gap [27,28]. The OD-ZnO particles absorb light at wavelengths lower than $\sim 390 \text{ nm}$ while 1D-ZnO particles lower than $\sim 400 \text{ nm}$ (Fig. 3b). The energy band gap estimated is 3.29 eV for OD-ZnO particles and 3.21 eV for 1D-ZnO particles (inset Fig. 3b). Considering that Kubelka-Munk treated data of diffuse reflectance are analogous to absorption results for bulk samples [27,28], the OD-ZnO particles exhibited a more intense absorption than 1D-ZnO.

3.2. PP/ZnO film characterization

In order to study the dispersion of ZnO particles in PP, the cryo-fractured surfaces at the highest concentration (PP/ODZnO-1.0 and PP/1DZnO-1.0) before xenon arc exposition were analyzed in SEM (Fig. 4). OD-ZnO particles (Fig. 4a) are well dispersed along the polymer fracture surface, however some of them form agglomerates reducing their high surface energy on account of their reduced dimensions. 1D-ZnO particles (Fig. 4b) are dispersed without a preferential orientation and retained the same morphology after melt processing.

The X-Ray patterns of pristine PP (Supplementary information Fig. S1) show four intense peaks at 2θ values of 14.1° , 16.9° , 18.5° , 21.3° related to (110), (040), (130) and (111) planes. These values indicate that the main crystallization phase of PP is the α -crystal form, since there is no evidence of the β -crystal form ($2\theta = 16.1^\circ$, corresponding to the (300) plane) [32]. The same diffraction peaks are also shown in PP/ZnO films with 1.0 wt% of OD-ZnO and 1D-ZnO particles (Supplementary information Fig. S1), suggesting that ZnO is not showing any influence on the crystal phase of PP. Furthermore, in the PP/1DZnO-1.0 sample, a peak at $2\theta = 31.8^\circ$ is attributed to ZnO, which it is difficult to determine in PP/ODZnO-1.0 sample due to the low concentration of OD-ZnO particles in PP (1.0 wt%) and the small size of this particle. For this reason, the Raman shift peak at 437 cm^{-1} [18] associated to E_2 phonon mode of ZnO is not clearly visible in Raman spectra (Supplementary information Fig. S2).

The Kubelka-Munk treated data of diffuse reflectance UV-spectra for pristine PP, PP/ODZnO-1.0 and PP/1DZnO-1.0 films are shown in Fig. S3 (Supplementary information). The Kubelka-Munk absorption curve of pristine PP film shows the peak absorption at 274 nm, which is related to the UV absorption of the commercial antioxidant used by the polymer manufacturer [3,33]. For the samples of PP/ZnO composite films, this antioxidant absorption peak is present in UV-spectra including the peak absorption of ZnO particles at 380 nm. The optical band gap values of the PP composite films (Supplementary information, Fig. S3 inset) are 3.24 eV and 3.18 eV for PP/ODZnO-1.0 and PP/1DZnO-1.0, respectively. These results clearly indicate that the optical properties of ZnO particles are not affected by their incorporation in PP.

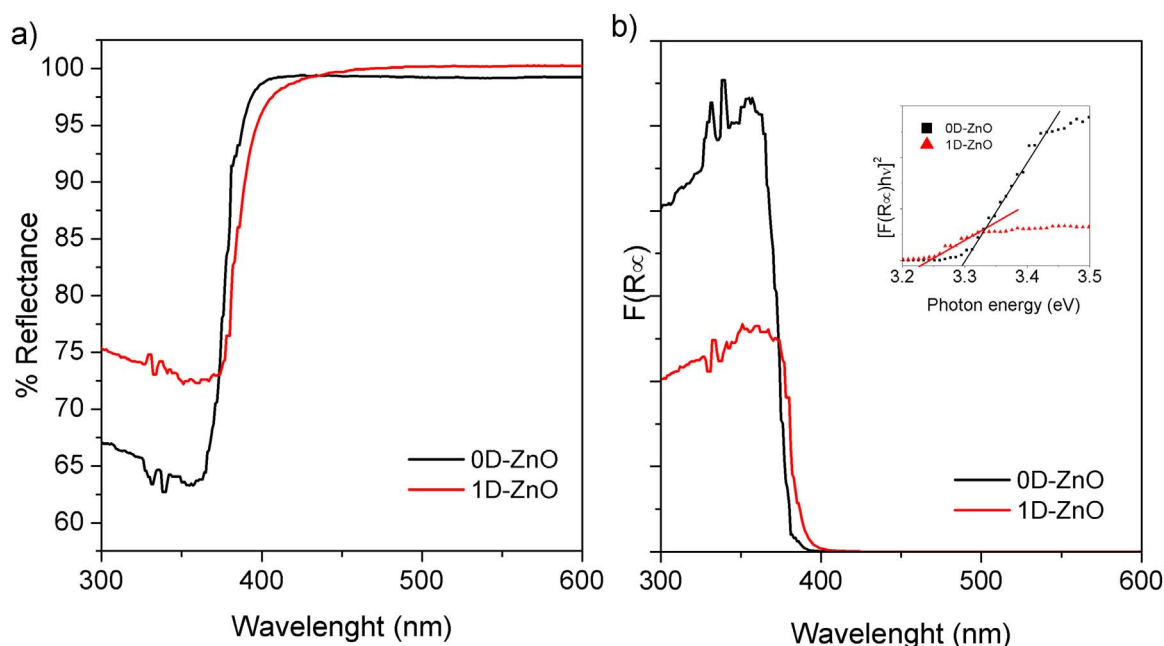


Fig. 3. UV-vis-DRS spectra of ZnO particles. a) Diffuse reflectance and b) Kubelka-Munk absorption curves (inset: Determination of band gap).

3.3. Photooxidation of PP/ZnO nanocomposites

At the beginning of irradiation exposure, all PP/ZnO nanocomposites films at different concentrations were translucent. As films were irradiated, the PP/1DZnO film retained its initial characteristics while the PP film and PP/0DZnO nanocomposites films showed loss of ductility and a change of color (Supplementary information Fig. S4). After 203 h of irradiation exposure, these films became brittle and cracked, and its handling was difficult because of their tendency to shatter. As a result of the film embrittlement, samples irradiated for more than 203 h were not analyzed by SEM and SEC in this work.

The change in weight loss of all PP/ZnO films under irradiation (Fig. 5) was measured. Only films with 0D-ZnO particles at all concentration, showed weight loss, being more significant at 0.75 and 1.0 wt% of nanoparticles. Comparing with the rest of the samples at 1.0 wt% of ZnO particles, the weight of PP and PP/1DZnO-1.0 films remained unchanged, whereas the PP/0DZnO-1.0 film decreased their values for the same time of exposure. These results indicate the progression of PP photodegradation to volatile products by means of the β -scission pathway, resulting in low molecular weight compounds [9,34,35] such as CO₂ and H₂O, as well as alkenes, alcohols, ketones, carboxylic acids, and lactones, whose formation and subsequent evaporation leads to the weight loss in the films.

Fig. 6 shows the SEM analysis on the film surface for pristine PP and

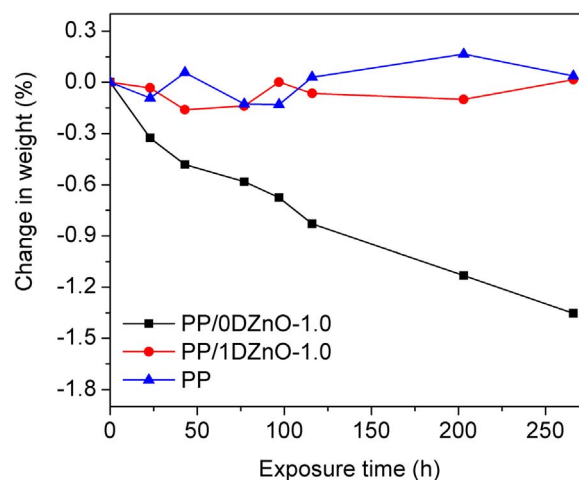


Fig. 5. Effect of morphology of ZnO nanostructure on the weight loss of PP/ZnO film nanocomposites during irradiation.

PP/ZnO composites with 1.0 wt% of ZnO after being exposed to irradiation for 203 h. PP/ZnO films after 203 h of exposure were difficult to handle because of the brittleness of the samples due to the high level of degradation, for this reason only those films could be observed by SEM.

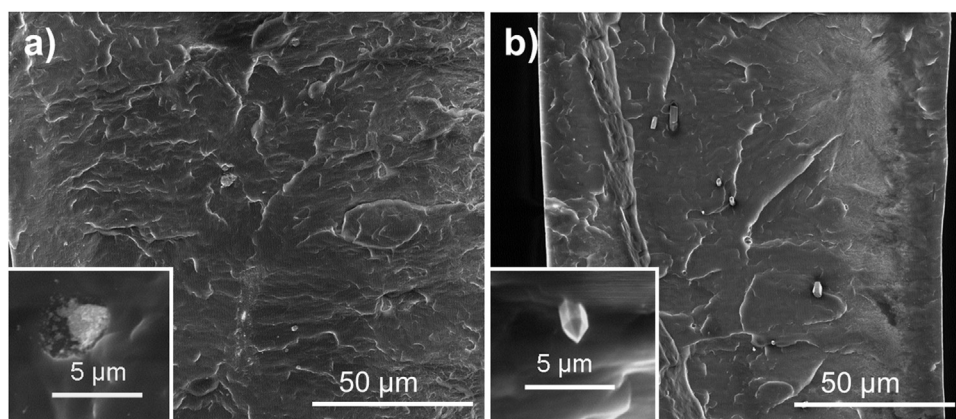


Fig. 4. SEM image of fractured surface of PP/ZnO nanocomposites films before irradiation at 1.0 wt% of ZnO a) 0D-ZnO, b) 1D-ZnO.

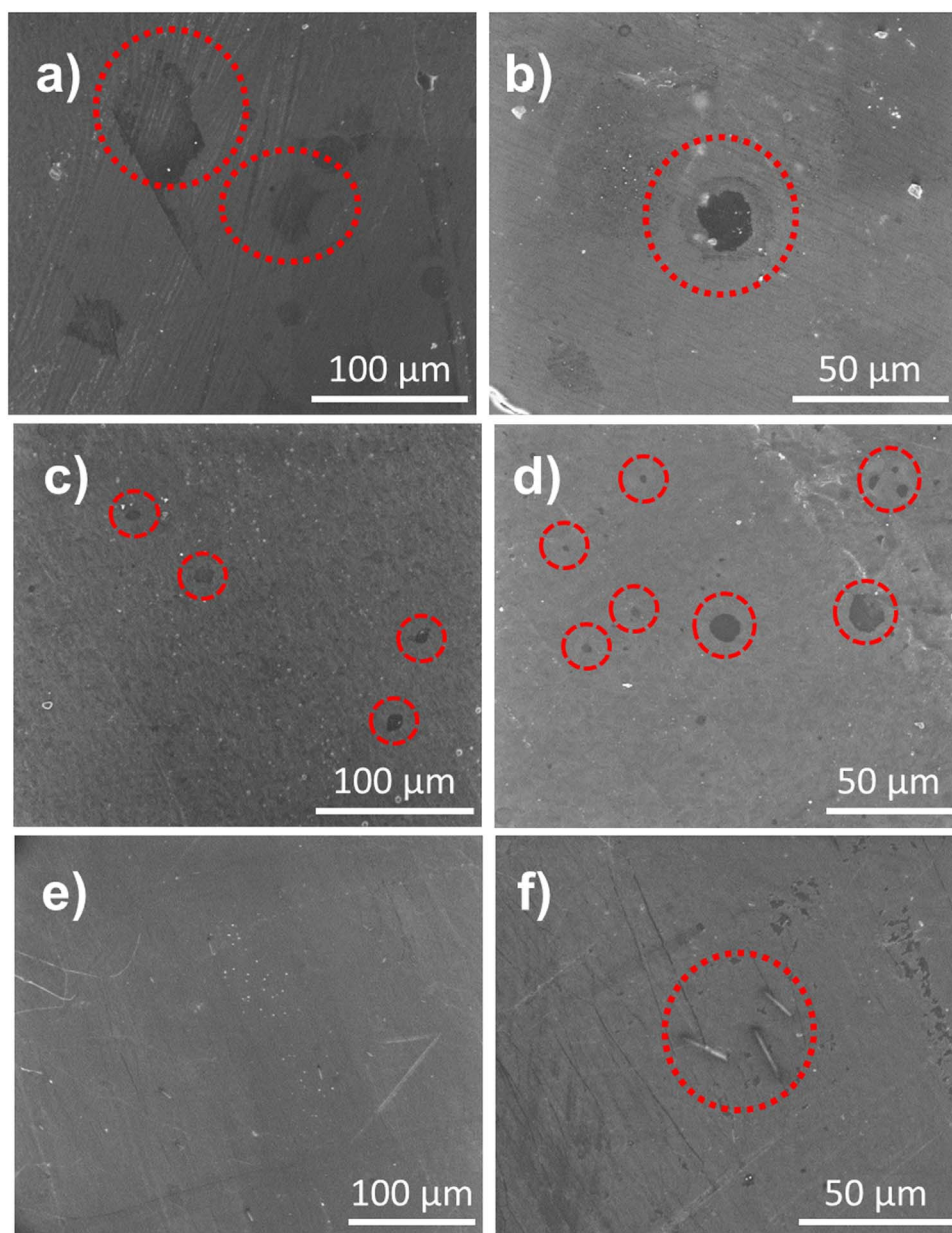


Fig. 6. SEM surface micrographs of (a and b) pristine PP, (c and d) PP/ 0DZnO-1.0 and (e and f) PP/ 1DZnO-1.0 films after 203 h of exposure.

The film surface of pristine PP after irradiation (Fig. 6a and b) shows some defects and small holes distributed along the surface, which is associated to a degradation process. Similarly, on the film surface of PP/0DZnO-1.0 (Fig. 6c and d) it is possible to observe regions with significant erosion, as well as pinholes of greater size and a more rugged surface, with a greater defect density than that of unloaded PP. The opposite case is observed for PP/1DZnO-1.0 (Fig. 6e and f), where large areas can be seen without any signs of degradation. These results support that 1D-ZnO nanostructures are acting as autocatalytic degradation inhibitors for PP.

During xenon arc irradiation all PP/ZnO nanocomposite films were periodically analyzed by FTIR spectroscopy in order to follow the photooxidation evolution. As consequence of PP photooxidation, the FTIR spectra exhibit absorption peaks of oxygenated functionalities such as hydroperoxides ($3600\text{--}3200\text{ cm}^{-1}$) and carbonyl species such as ketones, esters and carboxylic acids ($1800\text{--}1600\text{ cm}^{-1}$). The pristine PP film before irradiation (PP0) is used for comparative purposes (Fig. 7). The absorption peaks within the carbonyl region (1710 cm^{-1}) and hydroperoxide groups ($3600\text{--}3200\text{ cm}^{-1}$) for PP film exposed during 266 h (PP266) are more intense than PP0 as consequence of its

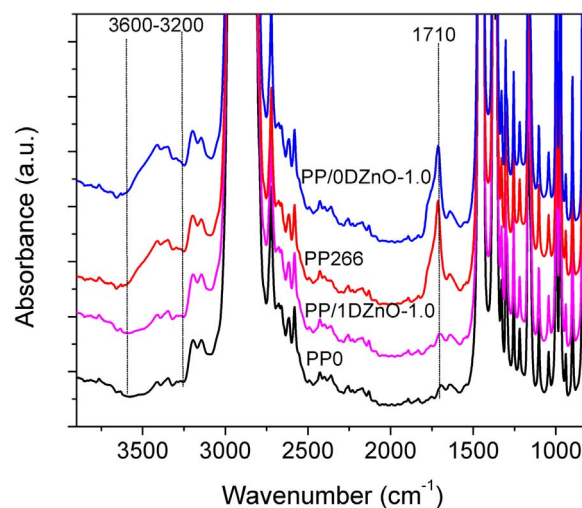


Fig. 7. FTIR spectra of films exposed during 266 h to xenon arc lamps: PP film (PP266), PP/0DZnO-1.0 and PP/1DZnO-1.0. Film of pristine PP (PP0) is included for comparison.

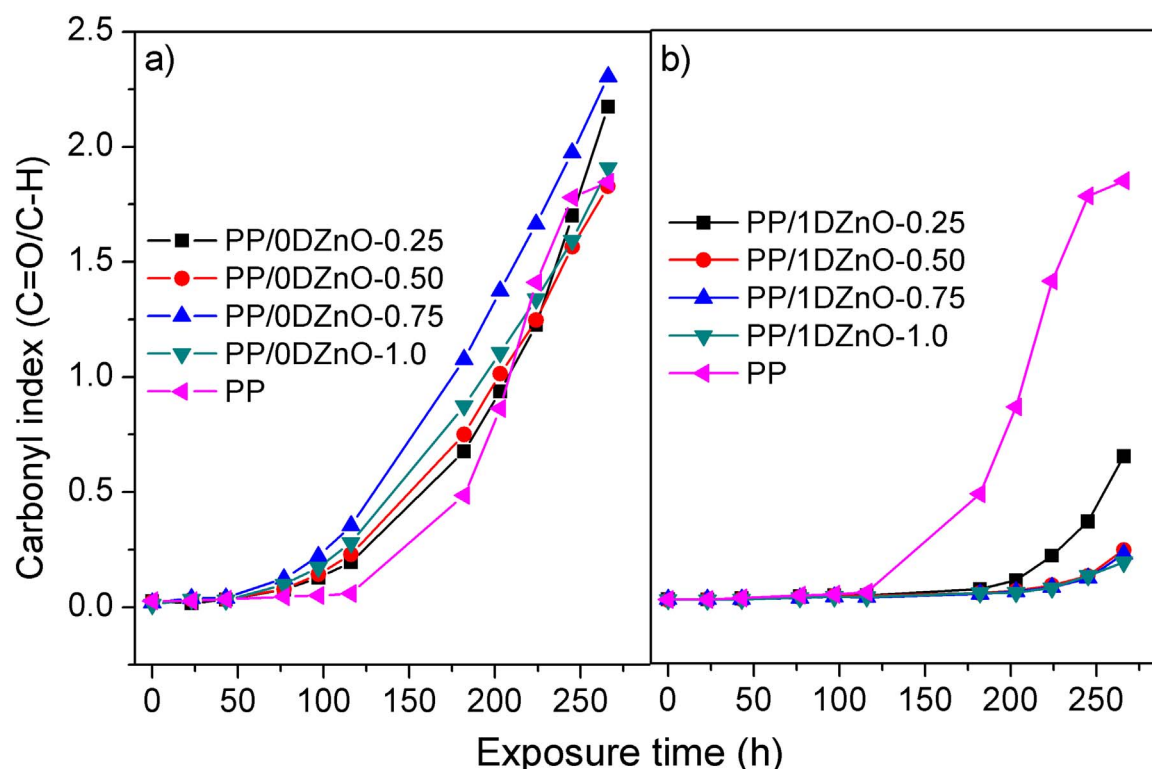


Fig. 8. Effect of a) ZnO-0D and b) ZnO-1D nanostructures on the carbonyl index at different irradiation times for PP/ZnO nanocomposites.

photooxidation. The development of this peak absorption is promoted by the auto-catalyzed oxidation of PP due to the presence of impurities such as metal traces from the polymerization reaction or high energy photo collision [12]. As shown for PP/0DZnO-1.0 in FTIR spectra (Fig. 7) after 266 h of exposure, the peak absorption at 1710 cm^{-1} is similar to that observed for PP266 after the same time of irradiation. The photooxidation mechanism for both PP266 and PP/0DZnO-1.0 could be similar, since the same additional absorption peak was observed in FTIR spectra. Whereas for the PP/1DZnO-1.0 film spectrum after 266 h of exposition, no absorption peak related to PP oxidation was observed, and it is very similar to pristine PP (PP0) spectrum.

Carbonyl index obtained from FTIR spectra is a critical parameter to evaluate the photooxidation mechanism and Fig. 8 shows the results for both PP/0DZnO and PP/1DZnO nanocomposites during exposition to xenon arc lamps. The carbonyl index for PP/0DZnO films (Fig. 8a) is higher as 0D-ZnO concentration increased from 0.25 to 0.75 wt%, however for the PP/0DZnO-1.0 film, the carbonyl index is close to PP/0DZnO-0.5 film. This suggests that the amount of chemical groups with carbonyl bonding are not increasing at higher content of 0.75 wt% 0D-ZnO nanoparticles as a consequence of the formation of ZnO agglomerates. The carbonyl index for PP/1DZnO films (Fig. 8b) showed a dramatic reduction compared to pristine PP, and it is similar for content higher than 0.25 wt% of 1D-ZnO. This implies that ZnO nanoparticles are stabilizing the PP macromolecules inhibiting the photodegradation process. The latter has also been reported on PP films using ZnO nanoparticles surface treated with organosilanes [12].

In addition, the photooxidation mechanism of PP is characterized by an induction period where oxygen uptake by the film occurs slowly, followed by a fast increase of photooxidation due to catalytic auto-oxidation of PP [34]. The induction time, in the plot of absorbance at 1710 cm^{-1} as a function of irradiation time, is the time when no oxidation is detected; after this time, the photooxidation rate increases reaching a constant value [33]. In our experiments, the induction time for photooxidation of pristine PP film (Fig. 8a) is around 115 h, while in PP/0DZnO nanocomposites is around 43 h for all concentration of 0D-ZnO, suggesting those nanoparticles increase the photooxidation

process.

There is a significant difference in the photocatalytic behavior of these ZnO nanostructures and their composites possibly due to a shape effect more than size effect, as it has been reported [22,24,36–41] in the decomposition of organic pollutants in aqueous solution. 1D-ZnO grew in [0001] plane during synthesis, which corresponded to the *c* axis of the hexagonal close packed cell of ZnO, so the (001) plane is only exposed in the prismatic termination of rods [26]. Considering the crystal face (001) as the plane with most energy and the most reactive surface since OH⁻ ions adsorb on this crystal plane [36], however, as 1D-ZnO grow in that direction decreasing the surface energy the nanostructure exhibits poor reactivity towards PP photooxidation reaction. Then, the photogenerated electrons and holes are more likely to be caught in the crystal structure defects or recombine before reaching the interface. The resulting effects make 1D-ZnO morphology capable of absorbing UV radiation on the film and whose structure would hinder the interaction of holes and electrons to form radicals. Furthermore, the photocatalytic behavior presented by 0D-ZnO is attributed to high surface energy due to the small particle size and the presence of the crystal plane (001) exposed on the surface. The increased presence of crystal plane (001) on the surface allows greater adsorption of hydroxyl groups, due to the positive charge on this plane which is associated to oxygen vacancies found on this plane [26]. This phenomenon allows the migration of photogenerated charge carriers to the active surface faster without dramatic recombination unlike 0D-ZnO producing free radicals that attack the macromolecule chain.

Fig. 9 shows the C1s spectra of PP, PP/0DZnO-1.0 and PP/1DZnO-1.0 samples after 203 h of irradiation. For comparison purposes, pristine PP film before irradiation is also included. In all cases the C1s peak was curve fitted in different peaks related to [42,43]: carbon of hydrocarbon (CH_n ~ 284.6 eV), carbon singly bonded to oxygen (C-O ~ 285.5 eV), carbon doubly bonded to oxygen (C=O ~ 286.7 eV), carbon double and single bonded to oxygen (O=C-O ~ 288.3 eV). The non-homogeneous charge peak (283.2 eV) is an artificial peak that is related to the non-homogeneous charging on the polymer surface caused by the insulating nature of the material. The first peak at 284.6 eV is related to

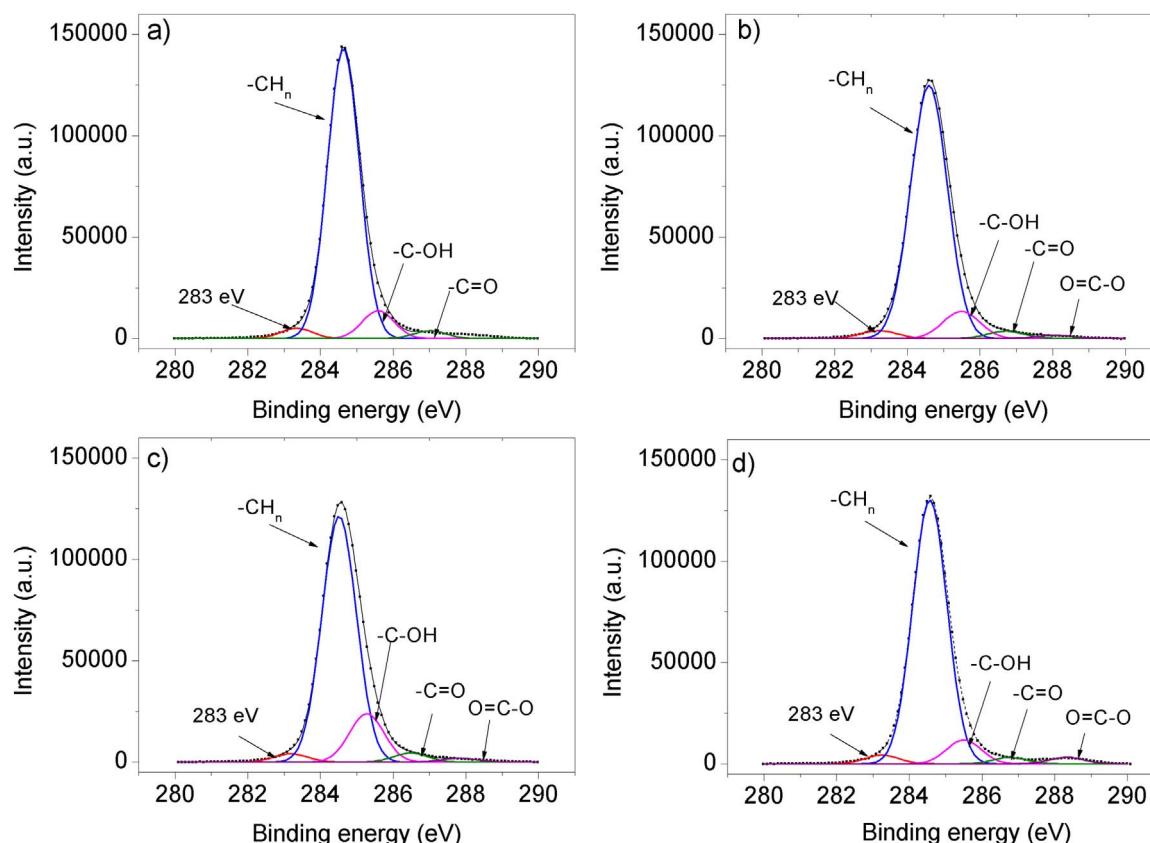


Fig. 9. C1s XPS spectra of surface exposed film PP/ZnO nanocomposites after 203 h of irradiation for b) PP, c) PP/0DZnO-1.0 and d) PP/1DZnO-1.0. For comparisons purposes a) PP film before xenon irradiation is shown.

the C-C and C-H bond of PP and the small shoulder peak between 285 and 289 eV represents the photooxidized surface groups after 203 h of irradiation. All samples show certain surface photooxidation degree, even pristine PP (Fig. 9a). However, the main difference after irradiation is the increased relative surface area of 285.5 eV (C-O) peak in PP/0DZnO-1.0 spectrum (Fig. 9c) compared to the rest of samples. This result indicates a higher concentration of C-OH groups supporting our findings about a photocatalytic behavior of 0D-ZnO particles.

The average molecular weight (Mn and Mw) and polydispersity of pristine PP films and nanocomposites with both ZnO particles containing 1.0 wt% after 203 h of exposure is presented in Table 2. It can be seen that after irradiation Mw of pristine PP decreased by 93.5%; while in the case of nanocomposite PP/0DZnO-1.0, it started decreasing to 93.7%. These values are quite similar since the photodegradation of PP/ZnO nanocomposites at 203 h of irradiation exposure is an auto-oxidation stage masking the contribution of 0D-ZnO (see Fig. 8). Furthermore, the Mw of PP nanocomposite PP/1DZnO-1.0 showed a decrease of 53% compared to pristine PP. These findings reinforce our previous observations about the behavior of ZnO particles as UV-screeners which was proposed by Lin et al. [14]. It is interesting to point out that in all cases, except for the PP/1DZnO-1.0, the polydispersity

decreased on average by about 50% compared with the pristine PP before irradiation, which shows the formation of shorter and homogeneous polymer chain length at 203 h of irradiation. In our experiments, the changes in average molecular weight data for irradiated PP/0DZnO-1.0 film were minimum compared to pristine PP after irradiation, however there was a slight change of mass loss (Fig. 5). The incorporation of 0D-ZnO could produce a greater amount of CO₂, as in the photodegradation of PE/ZnO nanocomposites [15], which is a consequence of the fact that the carbon atoms could be lost from the end of the polymer chain, causing a smaller reduction of average molecular weight. For this reason, a homogeneous dispersion and direct contact of ZnO to polymeric matrix to maximum the photocatalytic effect of 0D-ZnO particles is really necessary [15].

3.4. Mechanism of photooxidation of PP/ZnO nanocomposites

The incorporation of 0D-ZnO nanoparticles in the PP could accelerate the photooxidation mechanism due to that ZnO can be excited by the action of light source ($\lambda \leq 400$ nm) increasing the population of hydroxyl radicals ($\cdot\text{OH}$) [17–24,28,36–41]. In this process (Fig. 10), electrons from the valence band of ZnO are promoted to the conduction band resulting in the formation of holes (h_{vb}^+) and free electrons (e_{cb}^-) in their respective bands. If molecular oxygen and water molecules are in direct contact with the semiconductor surface, the reaction of e_{cb}^- with O₂ and h_{vb}^+ with H₂O produce the superoxide ion (O₂^{•−}) and hydroxyl radical (OH[•]), respectively. These reactive species are very important for the solid-phase photocatalytic degradation process; as they initiate the photodegradation by attacking the neighboring polymer chains, involving a hydrogen extraction from polymer backbone and the consequent formation of alkyl radicals. Those radicals quickly react with oxygen to yield peroxy radicals, and with other polymer chains to produce chain scissions. Consequently, peroxy radicals are capable of

Table 2

Molecular weight of PP with 1 wt% of 0D-ZnO and 1D-ZnO nanoparticles after irradiation during 203 h.

Sample	Mn (g/mol)	Mw (g/mol)	Polydispersity
PP (t = 0 h)	121,423	615,726	5.07
PP	15,736	39,715	2.52
PP/ 0DZnO-1.0	14,335	39,046	2.72
PP/ 1DZnO-1.0	70,429	289,383	4.10

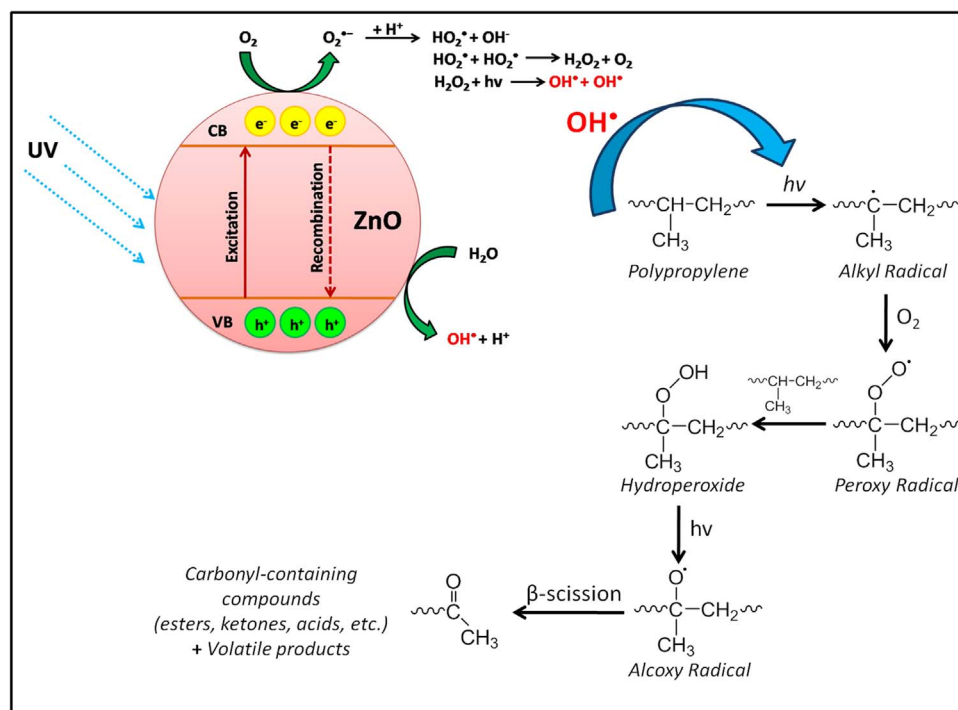


Fig. 10. Mechanism of photooxidation of PP in presence of ZnO particles.

stabilizing by extracting a hydrogen atom from a neighboring polymer chain becoming unstable hydroperoxide groups, which decomposed to produce alkoxy radicals and hydroxyl radicals by UV radiation. The alkoxy radicals can also promote scission in the polymer backbone or undergo β -scission. Finally, successive oxidation, generation of carbonylated products and reduction in molecular weight in polymer is observed in the polypropylene photooxidation mechanism [9,33]. In the case of 1D-ZnO particles, the xenon arc radiation is absorbed reducing the photooxidation in macromolecules, this absorbed light could be emitted as visible light [13].

4. Conclusions

The photooxidation of PP/ZnO nanocomposites by xenon-arc radiation was strongly influenced by morphology of ZnO nanoparticles and their concentration in the polymer. Quasi-spherical ZnO showed a photocatalytic effect for PP film, with increased carbonyl index due to the formation of oxidized functionalities (alcohol and carbonyl-containing moieties) and reducing its weight as exposure time to xenon-arc radiation increased. 1D-ZnO particles performed as a screening agent to xenon-arc radiation, inhibiting the autocatalytic oxidation of PP film and prevented any significant weight loss during the exposure. The obtained results showed the effect of ZnO morphology on the photooxidation process of PP/ZnO nanocomposites.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.mssp.2017.06.023>.

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